

Companion to Bird & de Moor, Algebra of Programming

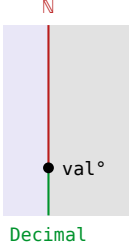
§6. Recursive Programs

§6.1. Digits of a number

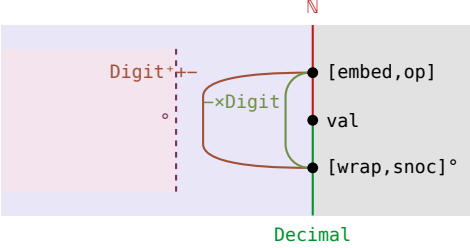
(6.1a)

$$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$$

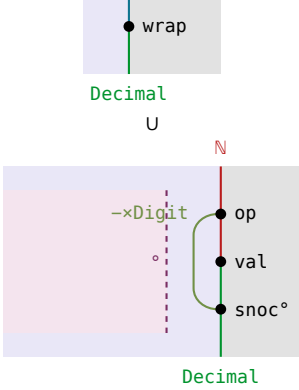
a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest



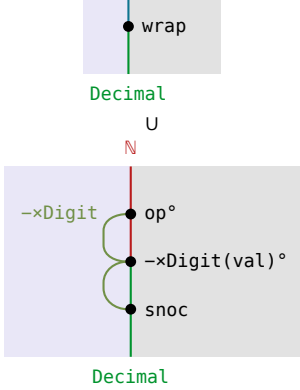
(a)



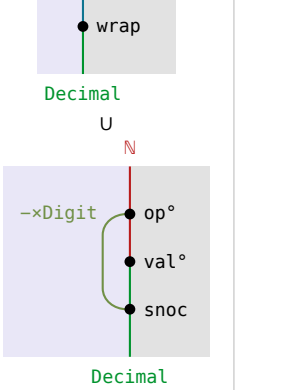
(b)



(c, d)



(e, f)



(g, h)

(a)	val°	the converse of val , which reads digits as a number
(b)	$([\text{wrap}, \text{snoc}]^\circ \text{Digit}^{++} - (-\times \text{Digit}(\text{val})) [\text{embed}, \text{op}])^\circ$	val undoes wrap or snoc , reads the front with val and combines by embed or op , so its converse runs that backwards
(c, d)	$(\text{wrap}^\circ \text{embed})^\circ \cup (\text{snoc}^\circ - \times \text{Digit}(\text{val}) \text{op})^\circ$	undoing wrap or snoc first gives two cases: a decimal and its last digit read by val then op , or one digit read by embed
(e, f)	$\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val})^\circ \text{snoc}$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then snoc ; or undo embed , then wrap
(g, h)	$\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$	undoing the read of the front while keeping the last digit is val° on the front alone

(6.1b)

$\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$
op° splits a number into its quotient and remainder by 10
$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$
op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
$(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$
embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

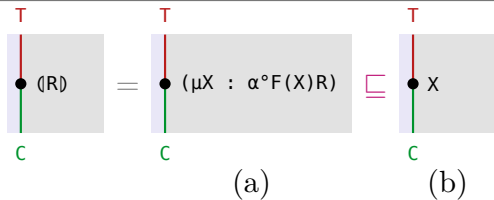
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

$$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$$

(6.2c)

a relation X that contains one unfolding of the fold's recursion at X contains the fold



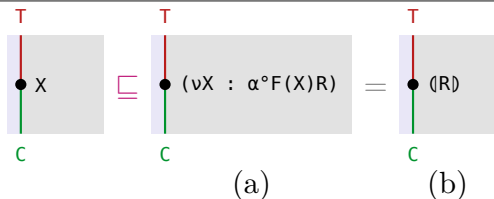
(a) $\langle R \rangle$ the least fixed point — Theorem 6.1, (6.2b)

(b) $\alpha^\circ F(X) R \sqsubseteq X$, μ below every prefix point

$$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$$

(6.2d)

a relation X contained in one unfolding of the fold's recursion at X is contained in the fold



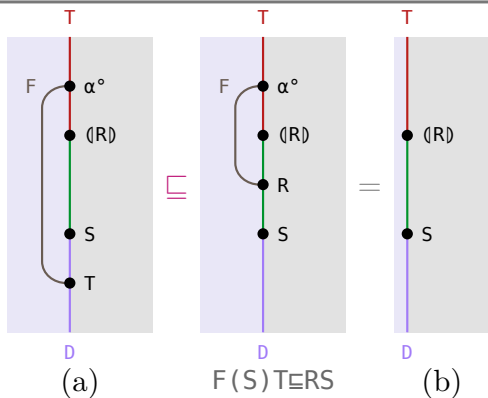
(a) $X \sqsubseteq \alpha^\circ F(X) R$, ν above every postfix point

(b) $\langle R \rangle$ the greatest fixed point

$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2e)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



(a) (6.2) at $X \triangleq \langle R \rangle S$

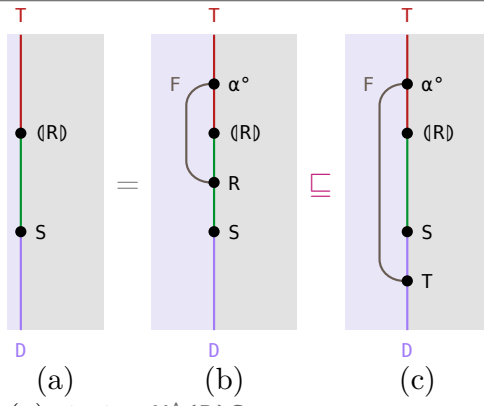
$F(RS) = F(R)F(S)$ — $\triangleq$ relator

(b) $F(\langle R \rangle)R = \alpha(\langle R \rangle)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce

$$RS \sqsubseteq F(S)T \Rightarrow (R)S \sqsubseteq (T)$$

(6.2f)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a) (6.3) at $X \triangleq (R)S$

(b) $\alpha(R) = F((R))R$, $\alpha^0 \alpha = 1$ — $\triangleq$ reduce

(c) $RS \sqsubseteq F(S)T$

$F(R)F(S) = F(RS)$ — $\triangleq$ relator

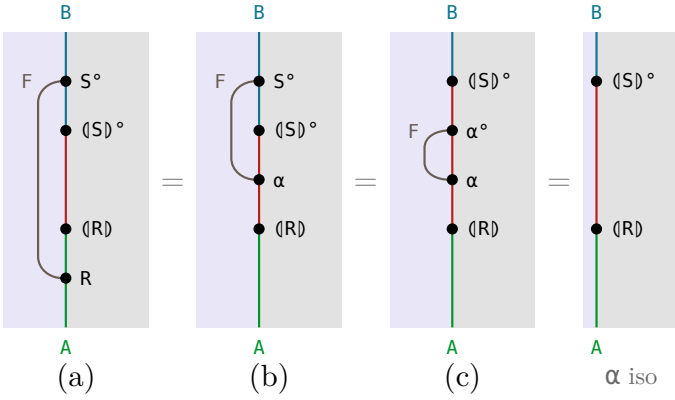
§6.3. Hylomorphisms

$$(\mathbb{S})^\circ \circ (\mathbb{R}) = (\mu X : S^\circ F(X)R) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

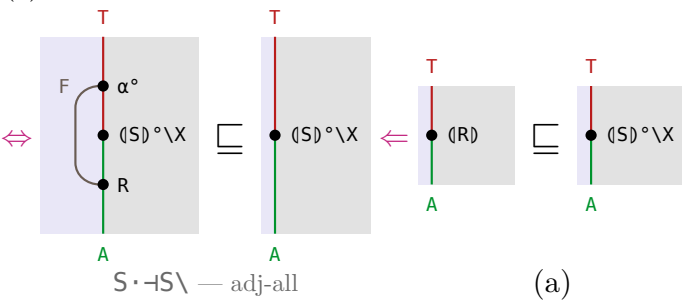
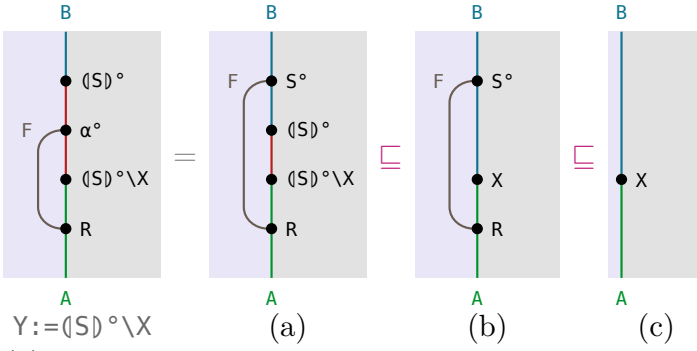
$$S^\circ F((\mathbb{S})^\circ \circ (\mathbb{R}))R = (\mathbb{S})^\circ \circ (\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ \circ (\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



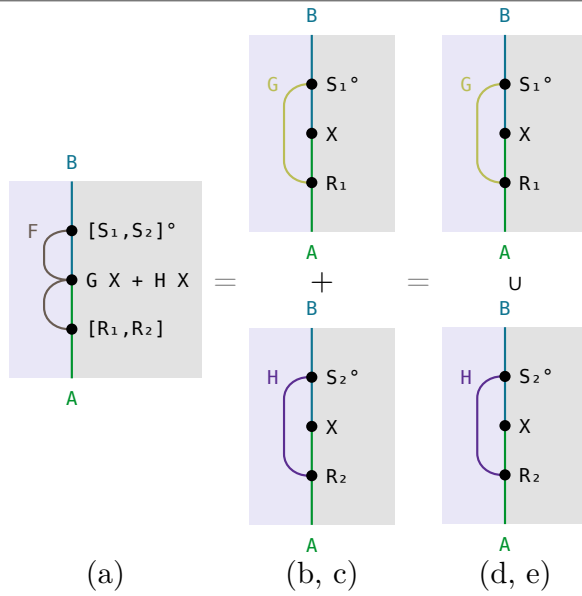
$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ \circ (\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ \circ (\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

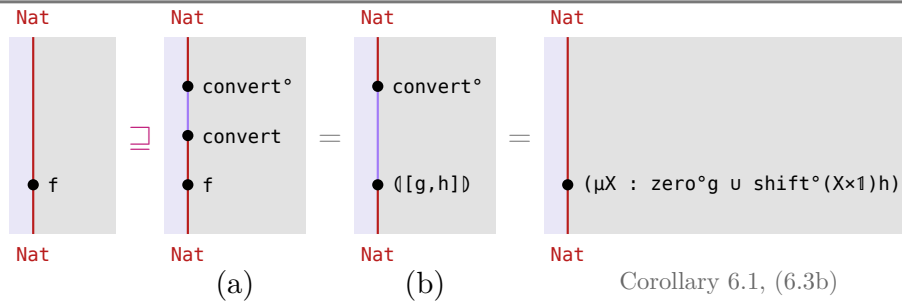


(a)	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	the body of Theorem 6.2 at $F(X) = G(X) + H(X)$
(b, c)	$[1, 1]^\circ ((S_1^\circ G(X) R_1) + (S_2^\circ H(X) R_2)) [1, 1]$	$[S, T] = (S + T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the + stands for the codiagonals ∇°, ∇ , which are not drawn
(d, e)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$\nabla^\circ (P + Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



- (a) convert simple
(b) fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of **a** to the power **n** and **d**

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

0 mod **b** is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

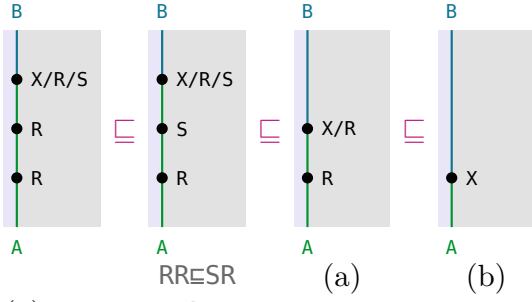
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and **d**

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R)$$

(6.5a)

R is inductive when $R \sqsubseteq SR$ for an inductive S

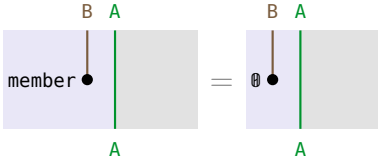


- (a) division by S
 (b) division by R

$$\text{member} = \emptyset$$

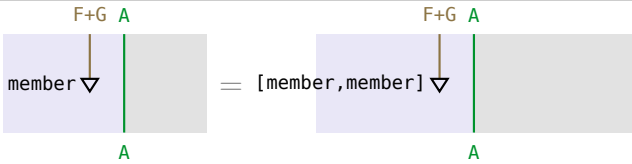
(6.5b)

a constant relator records no elements, so its membership is empty



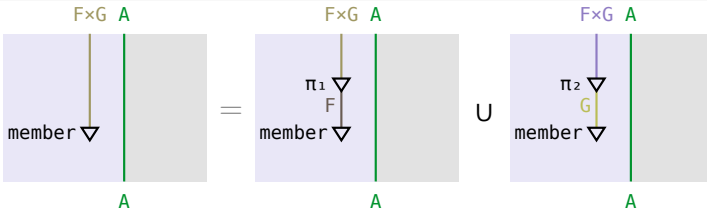
$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in



$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component



$$\text{inductive}(\text{Smember}) \wedge X = \text{SF}(X)R \wedge Y = \text{SF}(Y)R \Rightarrow X = Y$$

(6.5c)

when S **member** is inductive, two solutions X, Y of $X = \text{SF}(X)R$ are equal

$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge \text{SF}(X)R \sqsubseteq X \Rightarrow \text{Entire}(X)$$

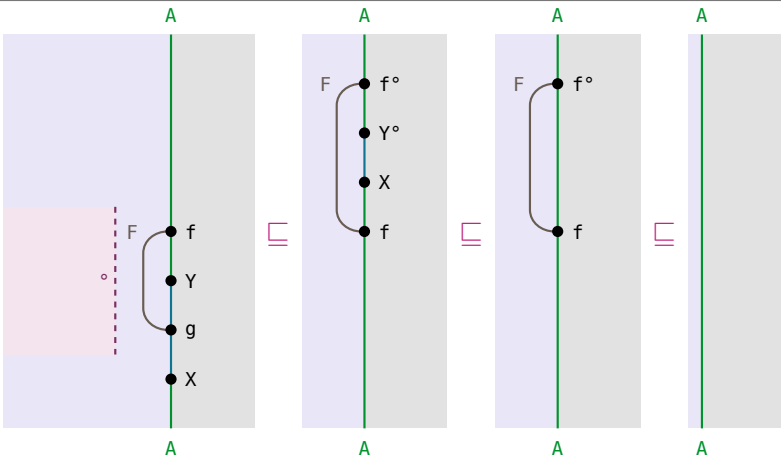
(6.5d)

when S **member** is inductive and S, R are entire, a pre-fixed point of $X \mapsto \text{SF}(X)R$ is entire

$$\text{inductive}(\text{gmember}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

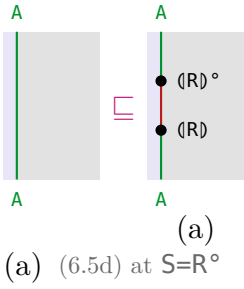
when g **member** is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map



- (a) $X = gF(X)f$, g simple

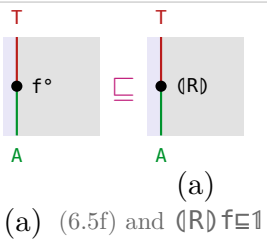
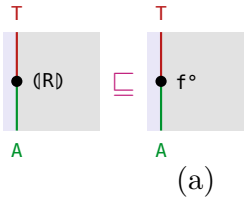
$$\text{inductive}(\mathbf{R}^\circ \text{member}) \wedge \mathbf{1} \sqsubseteq \mathbf{R}^\circ \mathbf{R} \Rightarrow \mathbf{1} \sqsubseteq (\mathbf{R})^\circ (\mathbf{R}) \quad (6.5f)$$

when $\mathbf{R}^\circ \text{member}$ is inductive and $\mathbf{R}$ is surjective, the fold $(\mathbf{R})$ is surjective



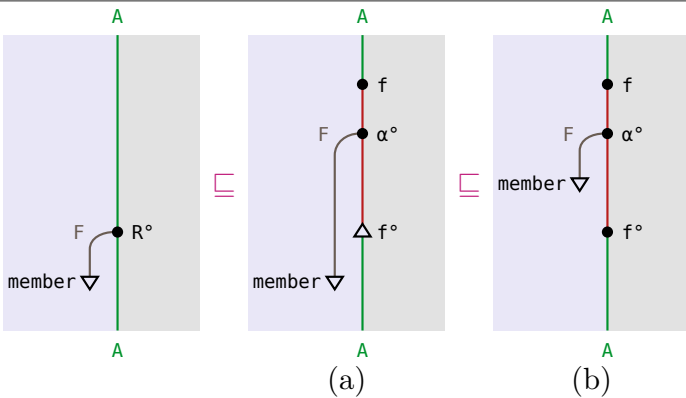
$$\text{Map}(\mathbf{f}) \wedge \mathbf{R} \mathbf{f} \sqsubseteq \mathbf{F}(\mathbf{f}) \alpha \wedge \mathbf{1} \sqsubseteq \mathbf{R}^\circ \mathbf{R} \wedge \text{inductive}(\alpha^\circ \text{member}) \Rightarrow \mathbf{f}^\circ = (\mathbf{R}) \quad (6.5g)$$

a map $\mathbf{f}$ with $\mathbf{R} \mathbf{f} \sqsubseteq \mathbf{F}(\mathbf{f}) \alpha$, for a surjective $\mathbf{R}$, has the fold $(\mathbf{R})$ as its converse



$$\text{Map}(\mathbf{f}) \wedge \mathbf{R} \mathbf{f} \sqsubseteq \mathbf{F}(\mathbf{f}) \alpha \Rightarrow \mathbf{R}^\circ \text{member} \sqsubseteq \mathbf{f} \alpha^\circ \text{member} \mathbf{f}^\circ \quad (6.5h)$$

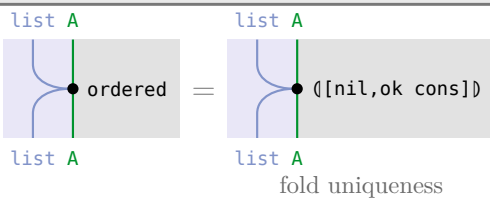
$\mathbf{R}^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by $\mathbf{f}$, so it is inductive when $\alpha^\circ \text{member}$ is



§6.6. Sorting by selection

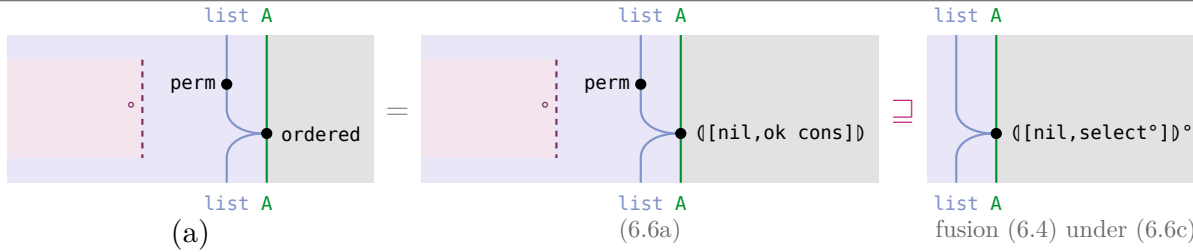
$$\text{ordered} = ([\text{nil}, \text{ok cons}]) \quad (6.6a)$$

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is $\mathbf{R}$ -below every element after it



$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

every output of unfolding the input by **select** is a sorted permutation of it

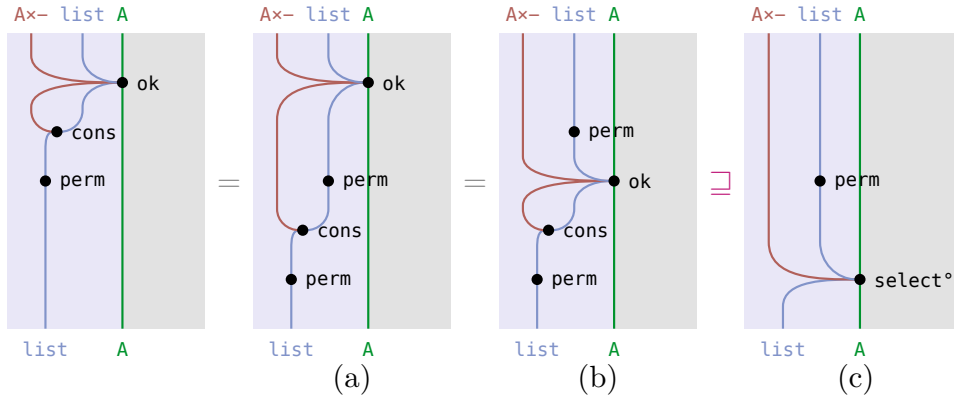


(a)

$\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



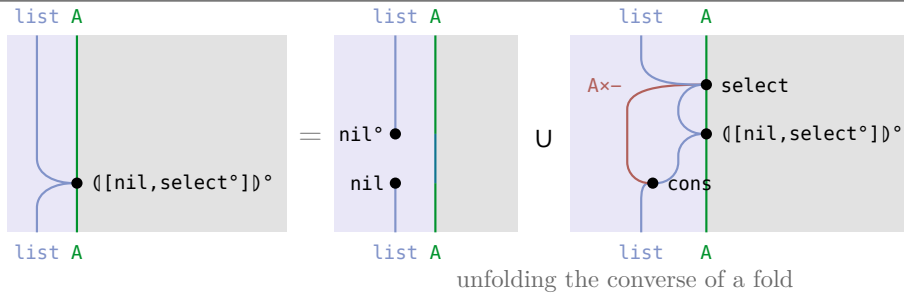
(a) $\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$

(b) Ex 6.22, $\text{ok}(1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$

(c) $\text{select}^\circ \sqsubseteq \text{ok cons perm}$

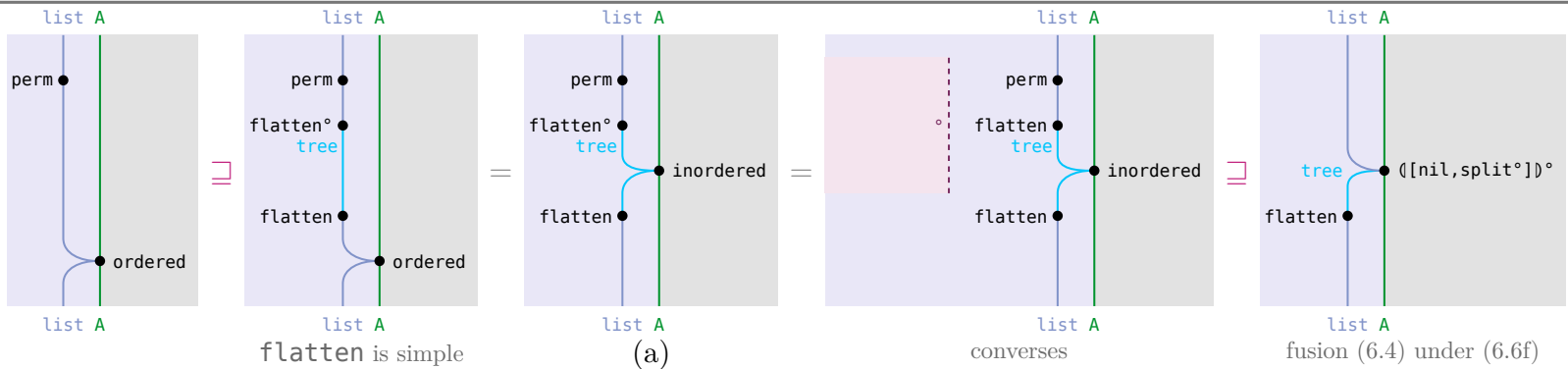
$$([\text{nil}, \text{select}^\circ])^\circ = \text{nil}^\circ \text{nil} \cup \text{select} (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

selection sort returns [] on [], and otherwise selects (a, y), sorts y and conses a back on



$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \sqsubseteq \text{flatten} \sqsubseteq \text{perm ordered} \quad (6.6e)$$

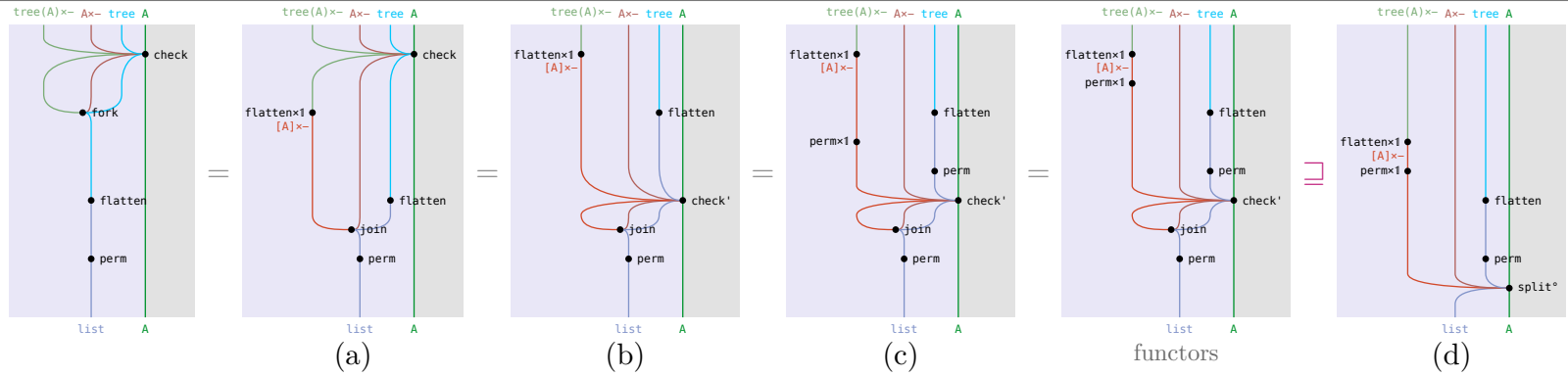
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



(a) $\text{flatten ordered} = \text{inordered flatten}$

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check fork flatten perm} \quad (6.6f)$$

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

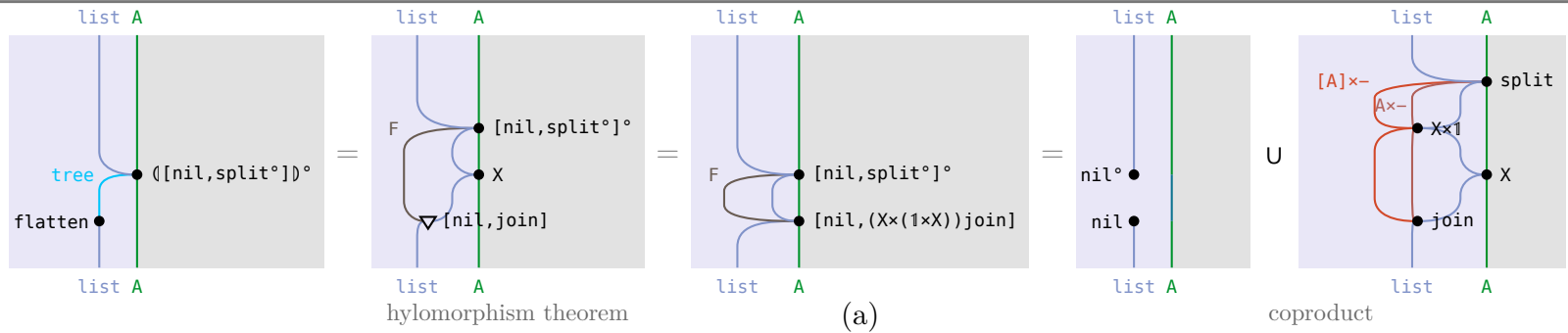


$$(\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{flatten} = \text{nil}^\circ \text{nil} \cup \text{split}$$

(6.6g)

$$(((\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{flatten}) \times (1 \times (\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{flatten}))) \text{join}$$

quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a

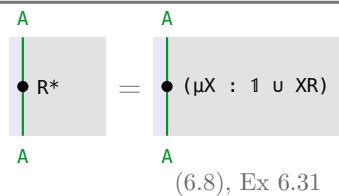


§6.7. Closure

$$R^* = (\mu X : 1 \cup XR)$$

(6.7a)

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***



$$1 \sqsubseteq X \wedge XX \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X$$

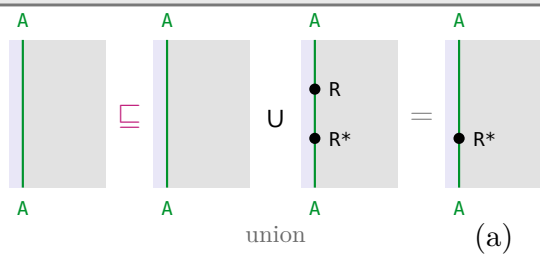
(6.7b)

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^*$$

(6.7c)

R* is reflexive



(6.7d)

$R \subseteq R^*$
 R^* contains R

identity
(a) $1 \subseteq R^*$ — (6.7c)
(b) fixed point

(6.7e)

$R^* R^* \subseteq R^*$
 R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

(a) composition distributes over $\cup$
(b) $RR^* \subseteq R^*$ — (6.7d)

(6.7f)

$1 \in X \wedge XX \subseteq X \wedge R \in X \Rightarrow R^* \in X$
a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

$1 \in X, R \in X$ $XX \subseteq X$

(6.7g)

$\text{inductive}(R) \wedge X = 1 \cup RX \Rightarrow X = R^*$
when R admits induction, R^* is the only X equal to $1 \cup RX$

(a) $(R^* \Rightarrow R^*/X) \cap R^* \subseteq R^*/X$

(6.7h)

$\frac{R^*}{3} = (\frac{1}{3}, \frac{RR^*}{3}) \text{ cup}$
the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors

(a) $\wedge(R \cup S) = \langle \wedge(R), \wedge(S) \rangle \text{ cup}$

$R - \emptyset = R$	(6.7i)
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away S $\cup$ T	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$		(6.7j)
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ		

C

$(\mu X : \varphi(\psi(X)))$

D

=

C

$\varphi(\mu Y : \psi(\varphi(Y)))$

D

Ex 6.35

$SR^*=(\mu X : S \cup XR)$		(6.7k)
S followed by any number of R steps is the least X containing S and closed under a further R step		

B

S

R*

A

=

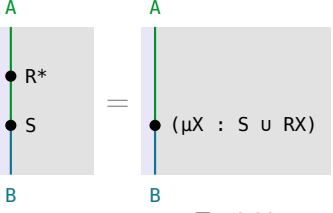
B

($\mu X : S \cup XR$)

A

p.160

$R^*S = (\mu X : S \cup RX)$
any number of R steps followed by S is the least X containing S and closed under an R step in front

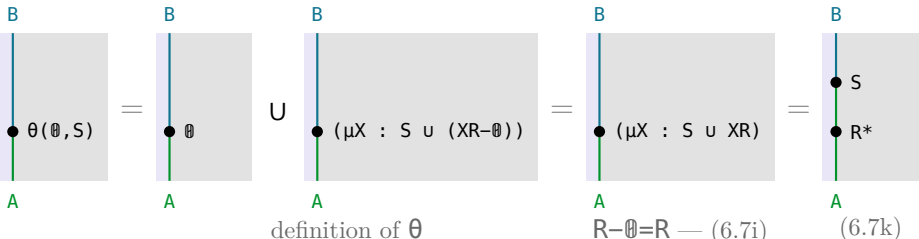


Ex 6.32

$\theta(\emptyset, S) = SR^*$

(6.7l)

started with nothing found and S to explore, θ computes SR^*



definition of θ

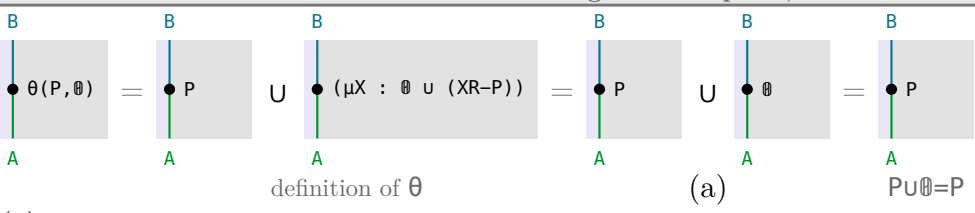
$R - \emptyset = R$ — (6.7i)

(6.7k)

$\theta(P, \emptyset) = P$

(6.7m)

with nothing left to explore, θ returns what it has found



definition of θ

(a)

$P \cup \emptyset = P$

(a) $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$

$\theta(P, Q) = \theta(P \cup Q, QR - P - Q)$ (6.7n)														
one round moves Q into the found part and explores the new R -successors of Q that are in neither														
$\begin{array}{c} B \\ \\ \bullet \theta(P, Q) \\ \\ A \end{array}$	=	$\begin{array}{c} B \\ \\ \bullet P \\ \\ A \end{array}$	$\cup$	$\begin{array}{c} B \\ \\ \bullet (\mu X : Q \cup (XR - P)) \\ \\ A \end{array}$	=	$\begin{array}{c} B \\ \\ \bullet P \\ \\ A \end{array}$	$\cup$	$\begin{array}{c} B \\ \\ \bullet (\mu X : Q \cup (XR - P - Q)) \\ \\ A \end{array}$	=	$\begin{array}{c} B \\ \\ \bullet P \\ \\ A \end{array}$	$\cup$	$\begin{array}{c} B \\ \\ \bullet Q \\ \\ A \end{array}$	$\cup$	$\begin{array}{c} B \\ \\ \bullet (\mu X : (Q \cup X)R - P - Q) \\ \\ A \end{array}$
				definition of θ					$QuS = Qu(S - Q)$ — (6.7i)					rolling — (6.7j)
=	$\begin{array}{c} B \\ \\ \bullet P \\ \\ A \end{array}$	$\cup$	$\begin{array}{c} B \\ \\ \bullet Q \\ \\ A \end{array}$	$\cup$	$\begin{array}{c} B \\ \\ \bullet (\mu X : (QR - P - Q) \cup (XR - (P \cup Q))) \\ \\ A \end{array}$	=	$\begin{array}{c} B \\ \\ \bullet \theta(P \cup Q, QR - P - Q) \\ \\ A \end{array}$							
		subtraction — (6.7i)								definition of θ				